Criterion C: Creating the Solution · Student Summative Answer Booklet

Ambassador International Academy, Dubai · Grade 9 Digital Design · MYP Year 4 · Data Science (DataSciensPro kit) · Sensor lesson: Light Detector (LDR)

Brief: In Criterion A you investigated a problem linked to the unit's UN Sustainable Development Goal. In Criterion B you designed a data-science solution with a chosen design (B3) and detailed plans (B4). In Criterion C you build the Light Detector (LDR) solution with the DataSciensPro kit, program it (Arduino sketch on the ESP32 and Python for the data), and test that it works as intended for your client. The sensor measures how much light reaches the sensor.

How to submit: Type your answers directly into this document. Use your own photographs, screenshots, serial output, charts and results, and paste images straight into the tables. Drawings may be on paper or digital. Save your work as you go and submit as your teacher instructs.

Name:		Class:	
Date:		Unit:	
Sensor lesson:	Light Detector (LDR)	Kit parts used:	

Criterion C checklist

- ☐ C1: A detailed, logical plan of the tasks and subtasks to build, code and test the solution, with the time and resources for each, so peers can follow it
- ☐ C2: A video or images that explain the code and the making process and show how they help solve the client's problem
- ☐ C3: A log for every lesson with images of the current development, ending with a solution that functions as intended
- ☐ C4: Images that compare the final solution with the B3 chosen design and the C1 plan, with every change fully justified

How Criterion C connects to Criterion B. Your solution is built from your Criterion B design. Use your B work in every strand.

Criterion B	Use it in Criterion C
B1 Design specification	Your test plan (C1) and final check (C4) test each success criterion.
B2 Developing ideas	Your chosen design from B3 is the starting point for your plan.
B3 Chosen design	Compare it with your final solution and code in C4.
B4 Planning drawings (circuit, flowchart, pseudocode)	Your Gantt tasks (C1) and code (C2) follow these. Explain any change in C4.

Your Criterion B work for CA3 · Daylight Tracker

Design brief. UAE school corridors and car parks often keep lights on all day because nobody measures daylight. Design a housing that holds the microcontroller and the light detector (LDR) so a school facilities team can read the daylight level.

B1 code	Success criterion from your specification
1.1	Aesthetics: pearl-white shell with sun-orange accents.
1.2	Aesthetics: edges filleted to at least 2 mm; cables hidden.
2.1	Cost: material and fabrication ≤ AED 80, excluding the kit Arduino and sensor.
2.2	Cost: reuses the school's Arduino and sensor module.
3.1	Customer: readable at a glance, no tools needed to read the result.
3.2	Customer: opens with one tool; USB cable reaches a laptop.
4.1	Environment: at least 80% recyclable PETG or reclaimed MDF.
4.2	Environment: printed and cut mass ≤ 300 g.
5.1	Size: envelope ≤ 140 × 120 × 200 mm.
5.2	Size: printed parts fit 200 × 200 mm; cut parts fit 300 × 300 mm.
6.1	Safety: smooth, UV-stable, no sharp edges, no exposed wires.
6.2	Safety: low-voltage USB power only.
7.1	Function: reports 0–100 light and separates day from night with fewer than 2 wrong readings per day.
7.2	Function: clear view of the sky over at least a 120° cone.
7.3	Function: reading available on the serial port.
7.4	Function: sensor swappable in under 2 minutes with hand tools.
8.1	Material: UV-stable PETG housing, translucent diffuser, 3 mm MDF base.
8.2	Material: flat parts laser-cut from MDF; complex shapes 3D-printed.

B3 chosen design	Concept 3, Daylight Sun-Tracker Post (34/40): a pole with a sky-facing head and a small solar panel. 130.0 × 110.0 × 189.8 mm, about 165 g, about AED 42.
B4 bill of materials	Base plate (3 mm MDF, laser-cut); base housing, pole, head arm and accent band (PETG, 3D-printed at 20% infill); solar panel; microcontroller with breadboard, LDR module and three status LEDs (kit parts); M3 × 12 bolt and nut sets (×10); cable-tie and wiring set.

B4 making worksheet. Turn these into timed tasks in your C1 plan. Number the tasks to match the steps.

Step	What you do
1	Check the CAD envelope against the limits.
2	Slice the printed parts in PETG (0.2 mm layers, 15–20% infill).
3	Laser-cut the MDF base plate.
4	De-burr and sand the printed parts.
5	Fit the LDR facing up through the diffuser.
6	Assemble following the exploded view.
7	Wire, upload a test sketch and check the Serial Monitor.
8	Install at the community location and confirm the view.
9	Function-test against specs 7.1 and 7.2.
10	Finish with paint or vinyl in the specified colours, then re-test.

About your Light Detector (LDR) kit

Parts	ESP32, LDR on GPIO 36 (VP), breadboard, jumper wires, optional LEDs on GPIO 25/26/27, USB cable
Pins	LDR on GPIO 36. The LDR is inverted: a low raw value means bright.
Arduino sketch	light-detection.ino reads analogRead(36), rescales it with map(raw, 650, 0, 0, 100) and prints a condition (DARK, DIM, BRIGHT, VERY BRIGHT) at 9600 baud.

Python scripts	1_data_collection_ldr.py (collect to CSV), 2_clean_analyse_ldr.py (clean and analyse), 3_visualize_ldr.py (charts).
Practice online	Lab 1 (virtual ESP32 + LDR) and Lab 2 (Python data lab) on the DS PRO site.

C1: Making Plans

Construct a logical plan, which describes the efficient use of time and resources, sufficient for peers to be able to follow to create the solution

Construct	Display information in a diagrammatic or logical form.
Develop	Improve incrementally, elaborate, or expand in detail. Evolve to a more advanced or effective state.

Level	Level Descriptor	Clarification
1-2	-	The plan is incomplete and not detailed. Each of the tasks and subtasks does not describe the time and resources required for its completion.
3-4	The student constructs a plan that contains some production details, resulting in peers having difficulty following the plan.	There is some evidence of planning to assemble, code and test the Light Detector (LDR) data-science solution, but the tasks listed are too simple, illogical, or would not result in a working solution.
5-6	The student constructs a logical plan, which considers time and resources, sufficient for peers to be able to follow to create the solution.	The plan is complete and makes sense in order to assemble, code and test the Light Detector (LDR) data-science solution. The time for each task is detailed alongside any resources required.
7-8	The student constructs a detailed and logical plan, which describes the efficient use of time and resources, sufficient for peers to be able to follow to create the solution.	The plan is complete and highly detailed, so a peer could assemble, code and test the same Light Detector (LDR) data-science solution. Each task and subtask fully describes the time and resources required.

The following questions can help you consider your planning:

- Does your plan contain a sequence of logical steps?
- Is the use of resources explained clearly?
- Have you considered alternative ways of creating the solution?
- Have you planned for testing at appropriate times in the making process?

Answer these in your plan
What are the major tasks (Task 1, 2, 3…) needed to build, code and test your solution?
For each major task, what are the subtasks, in order?
Mention the resources, learning time, development time, debugging time and testing time for each task.
Is the plan detailed enough for a peer to follow and make the same solution?

Your plan (Gantt chart). Write the dates of your lessons in the header. Shade each lesson column (e.g. yellow = learning, green = coding, blue = testing) where the task or subtask is done. Add or delete rows.

Task / Subtask	Description, resources, time (min)	L1 date:	L2 date:	L3 date:	L4 date:	L5 date:	L6 date:
Task 1: Make the housing (B4 steps 1–4)							
Subtask 1.1							
Subtask 1.2							
Task 2: Wire, code and collect data (B4							

step 7)							
Subtask 2.1							
Subtask 2.2							
Task 3: Assemble and install (B4 steps 5, 6, 8)							
Subtask 3.1							
Subtask 3.2							
Task 4: Test, debug and finish (B4 steps 9–10)							
Subtask 4.1							
Subtask 4.2							
Log entry with images (every lesson)							

Your test plan. Link each test to your Criterion B specification.

Test	What you measure	Method	Trials	Pass mark	B1 criterion
					1.1 / 1.2 look and finish
					5.1 size
					4.2 mass
					7.1 day and night readings
					7.2 sky view

C2: Creating the Solution

Demonstrate excellent technical skills when making the solution

Demonstrate	Make clear by reasoning or evidence, illustrating with examples or practical application.

Level	Level Descriptor	Clarification
1-2	The student demonstrates minimal technical skills when making the solution.	The student demonstrates minimal technical skills when making the Light Detector (LDR) data-science solution.
3-4	The student demonstrates satisfactory technical skills when making the solution.	The student demonstrates satisfactory technical skills when making the Light Detector (LDR) data-science solution.
5-6	The student demonstrates competent technical skills when making the solution.	The student demonstrates competent technical skills when making the Light Detector (LDR) data-science solution.
7-8	The student demonstrates excellent technical skills when making the solution.	The student demonstrates excellent technical skills when making the Light Detector (LDR) data-science solution.

Technical skills in the Light Detector (LDR) solution

Component	Excellent	Competent	Satisfactory	Minimal
Assembling and wiring	Assembled exactly as planned. Parts secure, wires tidy and short, the sensor is placed to measure what it should, and it can be taken apart for repair.	Assembled as planned. Parts are secure and wires are tidy.	Assembled with some help. Most parts are secure.	Assembled with a lot of help. Parts or wires are loose.
Sensor and Arduino sketch	Sketch is independent, commented and calibrated against real readings. Thresholds are justified with data, and the explanation covers every line.	Sketch reads the sensor correctly. Thresholds are chosen and explained.	Sketch adapted from the kit example with a basic explanation.	Sketch is incomplete or copied without explanation.
Python data code	Independent Python that collects, cleans (explaining what was removed and why), analyses and charts the data, with comments and a detailed explanation.	Python collects, cleans and charts the data. An explanation of the code.	Python adapted from the kit scripts with a basic explanation.	Python is incomplete or copied without explanation.
Testing and data	Planned tests are repeated in several conditions, results are recorded and the sketch is improved using the results.	Tests are carried out and results are recorded.	Some testing with few results.	Little or no testing.

What the kit expects from you: (1) Assembling and wiring: LDR on GPIO 36 with the LED extension, secure and tidy. (2) Sensor and Arduino sketch: analogRead, map(), constrain(), if / else if thresholds, Serial output at 9600 baud. (3) Python data code: parse a serial line, clean (range filter, z-score), moving average, charts. (4) Testing and data: readings in dark, shade, classroom lights and sunlight; thresholds improved with the data.

Accuracy of the solution: Have all components been included as detailed in the plan? Are the wires and connections made accurately?
Quality of the solution: Are the sketch and the Python code clear, commented and correct, and do the charts show attention to detail?
Assembly of the solution: Do all components fit together or combine as detailed in the plan?

Your evidence. Provide evidence (images) of the technical skills you used when creating the solution, and explain each code snippet: what it does, why you wrote it this way, and how you tested it.

Skill	Images, screenshots or data	Explanation
Assembling and wiring		
Sensor and Arduino sketch (explain it line by line)		
Python data code (explain it line by line)		
Testing and data		

Walkthrough video link (about 90 seconds: the circuit, the sketch, one serial output, a chart and how it helps the client):

C3: Following the Plan

Follow the plan to create the solution, which functions as intended

Create	Evolve from one's own thought or imagination, as a work or an invention.
Present	Offer for display, observation, examination, or consideration.

Level	Level Descriptor	Clarification
1-2	The student creates the solution, which functions poorly and is presented in an incomplete form.	The student creates the Light Detector (LDR) data-science solution, which functions poorly and is presented in an incomplete form.
3-4	The student creates the solution, which partially functions and is adequately presented.	The student creates the Light Detector (LDR) data-science solution, which partially functions and is presented adequately by maintaining a detailed lesson log.
5-6	The student creates the solution, which functions as intended and is presented appropriately.	The student creates the Light Detector (LDR) data-science solution, which functions as intended and is presented appropriately by maintaining a detailed lesson log while making it.
7-8	The student follows the plan to create the solution, which functions as intended and is presented appropriately.	The student follows the plan to create the Light Detector (LDR) data-science solution, which functions as intended and is presented appropriately by maintaining a detailed lesson log while making it.

Lesson log. Copy this table for every lesson (Lessons 1 to 6) and complete it at the end of the lesson.

Date:	Task: (header and content from your plan)
Goal: What did you plan to achieve in this lesson?	
Notes: What did you do? Paste images of your circuit, sketch, serial output, charts and tests. Mention problems and fixes.	
Reflection: What worked and what did not? What will you do next lesson? Did you follow your plan?	

(Copy the table above for each further lesson.)

Date:	Task: (header and content from your plan)
Goal: What did you plan to achieve in this lesson?	
Notes: What did you do? Paste images of your circuit, sketch, serial output, charts and tests. Mention problems and fixes.	
Reflection: What worked and what did not? What will you do next lesson? Did you follow your plan?	

C4: Changes to the Solution

Fully justify changes made to the chosen design and plan when making the solution

Describe	Give a detailed account or picture of a situation, event, pattern, or process.
Justify	Give valid reasons or evidence to support an answer or conclusion.
Outline	Give a brief account or summary.

Level	Level Descriptor	Clarification
1-2	-	The student is unable to justify the changes made to the Criterion B3 chosen design and the Criterion C1 plan when making the final solution in Criterion C3.
3-4	The student outlines changes made to the chosen design and plan when making the solution.	The student outlines the changes made to the B3 chosen design and the C1 plan when making the Light Detector (LDR) data-science solution.
5-6	The student describes changes made to the chosen design and plan when making the solution.	The student describes the changes made to the B3 chosen design and the C1 plan when making the Light Detector (LDR) data-science solution.
7-8	The student fully justifies changes made to the chosen design and plan when making the solution.	The student fully justifies the changes made to the B3 chosen design and the C1 plan when making the Light Detector (LDR) data-science solution, using images and test data.

Questions
Q1. Did you have to make changes to your original plan, sketch, Python code or circuit design? What changed?
Q2. Why did you make each change (test result, problem, data, client feedback)?
Q3. How did each change improve the solution for the client?

Justify your changes. Use images to compare your B3 chosen design with your final solution, sketch and charts, then complete the table. Use numbers from your data as evidence.

Original plan (B3 design / plan)	Final product	Reason for change
B3 chosen design: Daylight Sun-Tracker Post (130.0 × 110.0 × 189.8 mm, about 165 g)		
C1 plan: (write the original task or time)		

Final check against your B1 success criteria

B1 criterion	Test and result (use numbers)	Met?
5.1 Envelope ≤ 140 × 120 × 200 mm		☐ Yes ☐ Partly ☐ No
4.2 Mass ≤ 300 g		☐ Yes ☐ Partly ☐ No
2.1 Cost ≤ AED 80		☐ Yes ☐ Partly ☐ No
6.1 No sharp edges or exposed wires		☐ Yes ☐ Partly ☐ No
7.1 Fewer than 2 wrong readings per day		☐ Yes ☐ Partly ☐ No
7.2 Clear 120° sky view		☐ Yes ☐ Partly ☐ No

Self-assessment

Strand	Level I think I reached (1–8)	Evidence that supports it
C1 Plan		
C2 Technical skills		
C3 Following the plan		
C4 Changes		

Descriptors are taken from the school's Criterion C exemplar and the MYP Design guide. Check them against your teacher's current rubric.